

Question	Answer	Marks
11(a)	discrete amount/quantum/packet of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(b)	mostly dark/dark background	B1
	coloured lines	B1
11(c)(i)	6	A1
11(c)(ii)	1. maximum photon energy = $13.6 - 0.85$ $(= 12.75 \text{ eV})$	C1
	maximum kinetic energy = $(13.6 - 0.85) - 5.6$ $= 7.2 \text{ eV}$	A1
	2. energy = hc / λ	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / [(13.6 - 0.85) \times 1.60 \times 10^{-19}]$	C1
	$= 9.8 \times 10^{-8} \text{ m}$	A1

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Question	Answer	Marks
12(c)	fusion: two nuclei combine to form a (single) nucleus	B1